

Assignment-13

1 Solve $D^4 y + 16y = 0$

$$D^4 y + 16y = 0$$

$$m^4 + 16 = 0$$

$$m^2 = \pm 4i$$

$$m = \pm (1 \pm i)\sqrt{2}$$

$$y = e^{\sqrt{2}x} (C_1 \cos \sqrt{2}x + C_2 \sin \sqrt{2}x) + e^{-\sqrt{2}x} (C_3 \cos \sqrt{2}x + C_4 \sin \sqrt{2}x)$$

2 Write the mathematical structure of Legendre's diff eqn.

$$(1-x^2) \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} + n(n+1)y = 0$$

3 Define singular point

A point $x=a$ is singular if coefficients of DE become infinite or undefined there.

4 Find PI $D^2 y - 3Dy + 2y = e^{x+1}$

$$(D^2 - 3D + 2)y = e^{x+1}$$

$$(D-1)(D-2)y = e^{x+1}$$

$$y = Ae^x$$

$$PI = \frac{e^{x+1}}{(1-1)(1-2)} = -e^{x+1}$$

$$\boxed{y_p = -e^{x+1}}$$

Σ why series soln is needed for diff Eqn.

- DE has variable coefficients
- No closed-form solution exists
- Near singular points.

Section-B

1 Solve Cauchy Euler Eqn.

$$\cancel{xxxxxx} \quad x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + y = \log x \sin(\log x)$$

$$x = e^t \Rightarrow t = \log x$$

$$\frac{dy}{dx} = \frac{1}{x} \frac{dy}{dt} \quad ; \quad \frac{d^2 y}{dx^2} = \frac{1}{x^2} \left(\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right)$$

Substitute

$$x^2 y'' + xy' + y = \left(\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right) + \frac{dy}{dt} + y$$

$$= \frac{d^2 y}{dt^2} + y$$

$$\frac{d^2 y}{dt^2} + y = t \sin t$$

$$m^2 + 1 = 0$$

$$m = \pm i$$

$$y_c = C_1 \cos t + C_2 \sin t$$

PI

$$y_p = (At + B) \cos t + (Ct + D) \sin t$$

Substitute

$$y_p = -\frac{1}{4} t^2 \cos t + \frac{1}{4} t \sin t$$

$$t = \log x$$

$$y = C_1 \cos(\log x) + C_2 \sin(\log x) - \frac{1}{4} (\log x)^2 \cos(\log x) + \frac{1}{4} (\log x) \sin(\log x)$$

2 Solve the variation of Parameter
 $\frac{d^2 y}{dx^2} + y = \operatorname{cosec} x$

$$y'' + y = \csc x$$

CF

~~$$m^2 + 1 = 0$$~~

$$m = \pm i$$

$$y = C_1 \cos x + C_2 \sin x$$

PI

$$y_p = u_1 \cos x + u_2 \sin x$$

$$u_1' = \frac{-y_2 R}{w} = \frac{-\sin^0 \cdot \csc x}{1} = -1$$

$$N = \begin{vmatrix} \csc x \\ -\sin x \end{vmatrix}$$

$$u_2' = \frac{y_1 R}{w} = \frac{\cos x \cdot \csc x}{1} = \cot x$$

$$u_1 = -x$$

$$u_2 = \int \cot x dx$$

$$= \log(\sin x)$$

$$y = C_1 \cos x + C_2 \sin x - x \cos x + \sin x \log(\sin x)$$

3 Solve in series : $\frac{d^2 y}{dx^2} + x^2 y = 0$

$$y'' + x^2 y = 0$$

$$y = \sum_{n=0}^{\infty} a_n x^n$$

$$y'' = \sum n(n-1) a_n x^{n-2}$$

Substitute

$$\sum n(n-1) a_n x^{n-2} + x^2 \sum a_n x^n = 0$$

$$= \sum n(n-1) a_n x^{n-2} + \sum a_n x^{n+2} = 0$$

$$a_{n+4} = \frac{-a_n}{(n+4)(n+3)}$$

$$y_1 = a_0 \left(1 - \frac{x^4}{12} + \frac{x^8}{12 \cdot 7 \cdot 0} - \dots \right)$$

$$y_2 = a_1 \left(x - \frac{x^5}{20} + \frac{x^9}{20 \cdot 8 \cdot 9} - \dots \right)$$

$$\boxed{y = y_1 + y_2}$$

Section-C

1. Test the consistency of the following system of eqn. and if possible find the soln.

$$\begin{cases} x + y + z = 8 \\ 2x + 3y + 2z = 19 \\ 4x + 2y + 3z = 23 \end{cases}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 8 \\ 2 & 3 & 2 & 19 \\ 4 & 2 & 3 & 23 \end{array} \right]$$

$$R_2 \rightarrow R_2 - 2R_1$$

$$(0, 1, 0 | 3)$$

$$R_3 \rightarrow R_3 - 4R_1$$

$$(0, 2, -1 | -9)$$

$$R_3 \rightarrow R_3 + 2R_2$$

$$(0, 0, -1) - 3)$$

$$\underline{z = 3}$$

$$\underline{y = 3}$$

$$\underline{x = 2}$$

Ans

2 Solve the Bessel's Eqn

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 - n^2) y = 0$$

Std. Form

$$x^2 y'' + x y' + (x^2 - n^2) y = 0$$

$$y = \sum a_n x^{n+r}$$

substitute

$$r^2 = n^2 \Rightarrow r = \pm n$$

$$a_{k+2} = \frac{-a_k}{(k+r+2)^2 - n^2}$$

$$\boxed{y = C_1 J_n(x) + C_2 J_{-n}(x)}$$

solve in series

$$3x \frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} + y = 0$$

$$y = \sum_{n=0}^{\infty} a_n x^n$$

$$y' = \sum_{n=0}^{\infty} n a_n x^{n-1}$$

$$y'' + x^2 y = 0$$

$$\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} + x^2 \sum_{n=0}^{\infty} a_n x^n = 0$$

$$x^2 \sum a_n x^n = \sum a_n x^{n+2}$$

$$\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} + \sum_{n=0}^{\infty} a_n x^{n+2} = 0$$

let $n \rightarrow n+2$

$$\sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n + \sum_{n=0}^{\infty} a_{n-2} x^n = 0$$

$$a_{n+2} = \frac{-a_{n-2}}{(n+2)(n+1)}$$

Even

$$a_2 = 0 \quad a_4 = -\frac{a_0}{12}$$

$$a_8 = \frac{a_0}{12 \cdot 7 \cdot 8} \quad \dots$$

$$y_1 = a_0 \left(1 - \frac{x^4}{12} + \frac{x^8}{12 \cdot 7 \cdot 8} - \dots \right)$$

Odd

$$a_3 = 0 \quad a_5 = -\frac{a_1}{20}$$

$$a_9 = \frac{a_1}{20 \cdot 8 \cdot 9}$$

$$y_2 = a_1 \left(x - \frac{x^5}{20} + \frac{x^9}{20 \cdot 8 \cdot 9} - \dots \right)$$

$$y = a_0 \left(1 - \frac{x^4}{12} + \frac{x^8}{12 \cdot 7 \cdot 8} - \dots \right) + a_1 \left(x - \frac{x^5}{20} + \frac{x^9}{20 \cdot 8 \cdot 9} - \dots \right)$$

Ans